GAO Ziming

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Research Interests

Human-computer interaction; human-AI interaction; context-aware assistance for physical activities; physical skill learning; wearable and multimodal interaction.

Educational Background

The University of Hong Kong 09/2026-present

MSc(Eng) in Innovative Design and Technology; expected completion: 07/2027

Beihang University (BUAA) 09/2021-06/2025

BEng in Industrial Design

- CGPA: 3.37/4.0; Major GPA: 3.67/4.0; Last-two-year GPA: 3.63/4.0
- BUAA Scholarships: Top Prize of Encouragement Scholarship (top 5%) & Second Prize of Merit Scholarship (2023/24); Second Prize of Encouragement Scholarship (2022/23)

Research Experience

IDEA Lab, Zhejiang University 03/2026-present

Research Intern; research mentors: Lingyun Sun and Hongbo Zhang

Focus: Human-AI interaction and context-aware assistance for physical activities.

- Presented the group's research on contextual AI assistance at the IDEA Lab Summer Meeting, Zhejiang University, on 31 July 2026.

Physical-Knowledge Grounding

Ongoing research; preparing a manuscript for CHI submission.

- Responsible for the formative study on adapting online physical-skill tutorials to users' materials, tools, and environments through questionnaire and interview research.
- Synthesized findings into research challenges and design goals; created research figures and supported experimental activities.

Situation-Bound Intent

Manuscript submitted to ACM TOCHI; fourth author.

- Designed modular wearable hardware for the application-space component of a system for situation-aware prospective memory assistance.
- Created user-study result figures and refined other manuscript visuals.

School of Mechanical Engineering & Automation, BUAA 09/2025-03/2026

Part-time Research Assistant, High-temperature and High-pressure Flexible Forming Laboratory

- Performed Abaqus simulations of component forming; analyzed and visualized simulation results.

Publication

Gao, Z., & Xu, W. (2026). Product innovation design based on the sail-shaped structural characteristics of Velella. Journal of Northwest Normal University (Natural Science), 62(4), 92–100. In Chinese. DOI: 10.16783/j.cnki.nwnuz.2026.04.009.

Research Presentation

Contextual AI Assistance in Physical Activities. Presented on behalf of the research group at the IDEA Lab Summer Meeting, Zhejiang University, 31 July 2026.

Selected Projects and Awards

VelellaNexus — Undergraduate Graduation Project

- Developed a biomimetic modular floating transport platform through morphology research, physical experiments, simulation, and prototyping; published a first-authored journal article.
- NCDA 2025: National Second Prize and Beijing First Prize;
  First Prize, 13th Beijing Universities Industrial Design Competition.

Skills and Teaching Experience

- Research: Formative studies, questionnaire and interview synthesis, research visualization.
- Tools: Figma, Arduino, Processing, Rhino, SolidWorks, KeyShot, Illustrator and Photoshop.
- Language: Mandarin Chinese (Native); English (IELTS 6.5); Japanese (JLPT N2)
- Teaching Assistant, BUAA: Product Design (Intelligent Interaction), Interactive Programming Techniques, and Illustration.